RACHNA KADAM

Brooklyn, NY | +1 (929) 822-3163

rnk9706@nyu.edu | www.linkedin.com/in/rachna-kadam-b348891a7/ | Github | Portfolio

SUMMARY

Mechatronics and Robotics engineer with hands on experience in embedded hardware design, sensor integration, and vision based control for industrial robotic systems.

SKILLS

- **Hardware:** PCB Design, Embedded systems, KiCad, signal conditioning, relay/high-voltage switching, motor driver integration
- **Bench Instruments:** Oscilloscope, Logic Analyzer, Function Generator, Multimeter, Power Supply; used for embedded circuit debugging
- **Languages:** Embedded C, Python, C++, MATLAB
- **Embedded Systems:** ESP32, Arduino, Raspberry Pi, Jetson; I2C, UART, SPI, PLC Programming (IndraWorks – Bosch Rexroth), SCADA/HMI
- **Sensors & Signal Acquisition:** LiDAR, IMU, Quadrature/Magnetic Encoders, Current/Voltage, Ultrasonic, IR, Temperature, Monocular/Stereo Camera; ADC/DAC
- **Controls:** PID, LQR, SMC, System Identification, State Estimation, Model-Based Design, Real-Time Feedback Control, Tustin Discretization
- **Signal Processing & Perception:** Kalman Filter, Complementary Filter, Sensor Fusion, OpenCV, Numerical Differentiation
- **Tools & CAD/CAE:** Git/GitHub, Linux (Ubuntu), SolidWorks, Fusion 360, Ansys, MATLAB-Simulink, 3D Printing, Laser Cutting

EXPERIENCE

Robotics Hardware Intern – FANUC Industrial Robot (Grinding) **Apr - May 2026**

New York University, Tandon School of Engineering

- Programmed and integrated a FANUC industrial robot with magnetic grippers for automated grinding, developing and validating sensor-actuator sequences for joint positioning that achieved sub-millimeter placement accuracy across production runs; designed gripper mount & robot base mount in SolidWorks for secure fixturing.
- Built a full camera calibration pipeline for the robot workspace — intrinsic calibration with a ChArUco board, extrinsic calibration with ArUco markers, and eye-in-hand & hand-eye calibration for an end-effector mounted camera.
- Integrated an Intel RealSense RGBD camera into the perception pipeline for task planning, pick-and-place, and evaluated VLM-based approaches for object detection and pose estimation.

Research Assistant – NYU Mechatronics, Controls and Robotics Lab **June 2025 – May 2026**

VLM–LLM Integrated Multi-Robot Embedded System | Dr. Vikram Kapila

- Designed an interrupt-driven, 4-protocol wireless communication bridge (Wi-Fi, BLE, Classic BT, ESP-NOW) on ESP32, with hardware-level mode-switching via 5s button hold.
- Built a custom binary encoding layer compressing JSON payloads from 55 bytes → 7 bytes (87% reduction), sustaining throughput across scaled multi-robot deployments.
- Validated firmware across 4 heterogeneous MCU architectures – ATmega328P, ESP32-S2, ESP32-S3, RP2040 W – achieving zero packet loss under concurrent multi-robot coordination.
- Integrated bridge into a full autonomy stack - RRT* planning → Pure Pursuit + PID, achieving <50ms command latency from LLM to hardware execution.

Course Assistant – Automatic Controls Laboratory **Sep 2025 – May 2026**

New York University, Tandon School of Engineering

- Conducted and redesigned controls lab experiments (DC motor, maglev, inverted pendulum, water tank) for 50+ students using Quanser/MATLAB-Simulink, cutting setup time 60% via a custom Python GUI and Arduino-deployed embedded C controllers.

Mechanical Design & Validation Engineer (SAE BAJA) **Sep 2021 – May 2022**

MSPTME Racing Team

- Designed an ATV hydraulic braking system in Fusion360 CAD, achieving 15% weight reduction validated across 40+ dynamic test runs.
- Ran FEA (Ansys) stress analysis with 96% correlation to field results, securing 10th place in SAE BAJA Phase 2.

PROJECTS

Mecanum Drive Embedded Control System **Feb 2026 – April 2026**

- Designed a 4-wheel mecanum drive embedded system on ESP32-S3, interfacing 4x AS5600 magnetic encoders, MPU9250 9-axis IMU, and 12V battery monitoring via 12-bit ADC voltage divider, achieving 95% sensor read reliability at 100Hz.
- Implemented a 3-tier ROS2 architecture - ESP32-S3 → Raspberry Pi 3B → laptop - reducing command latency by 35% over single-board alternatives; used oscilloscope and logic analyzer to debug I2C/SPI sensor communication during bring-up.
- Deployed SLAM Toolbox + Nav2 on RPLidar A1M8 data, fusing wheel odometry + IMU via Complementary + Kalman Filter, improving localization accuracy by 40% over odometry alone, visualized in RViz.
- Deployed Sliding Mode Controller (SMC) for robust per-wheel motor control, reducing velocity tracking error by 28% over standard PID.

Motion Signal Acquisition & Fusion System: Punch n Pop **April 2025 – May 2025**

- Designed a multimodal signal acquisition system using dual MPU6050 IMUs sampled at 200Hz, computing jerk derivatives in real time for 95% gesture detection reliability.
- Built a sensor fusion pipeline on Raspberry Pi 400 combining IMU data with OpenCV HSV-based optical tracking, achieving sub-100ms end-to-end signal-to-output latency.

Pneumatic Actuated Parking System **Oct 2023 – Nov 2023**

- Designed and wired a full sensor-actuator pipeline on Arduino Mega 2560 – IR + dual HC-SR04 ultrasonic sensors → SG90 servo → 5/2 DCV-controlled double-acting pneumatic cylinder – with distance-threshold actuation logic.
- Engineered relay-based high-voltage switching (220V AC) via 2-channel SRD-05VDC relay module, isolating MCU logic from actuator power rail with zero false triggers across all test runs.

EDUCATION

New York University, New York, NY | *Master of Science, Mechatronics and Robotics (GPA: 3.7)* **May 2026**

NMIMS University, Mumbai, India | *Bachelor of Technology, Mechatronics (GPA: 3.69)* **May 2024**